

Quality-Inspection — Standard Operating Procedure

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Quality-Inspection — Standard Operating Procedure

Audience: Quality Worker · Maintenance Tech · Line Manager
Applies to: Line-B, station 4, Quality-Inspection

1. Purpose

Quality-Inspection is the end-of-line gate for Line-B. A multi-camera vision cell checks the trimmed housing for dimensional conformance, surface defects, and missing features before the part is released for shipment. Reject decisions follow `Quality_Reject_Disposition_Policy.pdf`.

2. Normal operating envelope

- Ideal cycle time: **20 s**
- Max acceptable cycle time: **26 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.2 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning here	First check
Starved	Trimming-Station (station 3) is not feeding parts.	Verify Trimming state — and back-check Hydraulic-Press at station 2.
Blocked	Discharge conveyor is full.	Verify the part-out conveyor.

4. Faults & corrective actions

4.1 Camera-Fault

- **Symptoms in telemetry:** `fault_type = "Camera-Fault"`. Sudden spike in rejects with vision-system reason codes, or no image returned at all.

- **Likely root cause:** camera lens fouled, focus drifted, or camera controller communication dropped.
- **Corrective action:**
 1. Wipe lens; check for debris on the optical path.
 2. Cycle camera power; verify image acquisition in the diagnostic tool.
 3. If degraded, replace the camera; restore calibration from the saved recipe.
- **Restart criteria:** master part (known-good) passes inspection three times in a row.

4.2 Sensor-Drift

- **Symptoms in telemetry:** fault_type = "Sensor-Drift". Rising false- reject rate against parts that pass downstream verification.
- **Likely root cause:** distance / proximity sensor calibration drift, or temperature-induced gain change.
- **Corrective action:**
 1. Run the calibration routine against the master gauge block.
 2. If still out of tolerance, replace the sensor.
 3. Re-verify all measurement features against the recipe.
- **Restart criteria:** calibration error within ± 0.02 mm.

4.3 Light-Failure

- **Symptoms in telemetry:** fault_type = "Light-Failure". Vision system rejects increase; image histograms show low contrast.
- **Likely root cause:** ring-light LED failure or driver fault.
- **Corrective action:**
 1. LOTO and replace the failed light bar.
 2. Restore brightness setpoint per recipe.
 3. Re-validate against the master part.
- **Restart criteria:** master part image meets contrast threshold.

5. Preventive maintenance schedule

- **Daily:** wipe lenses; run master-part verification at shift start (see Shift_Handover_Checklist.pdf).
- **Weekly:** clean diffusers; check light driver currents.
- **Monthly:** full recipe revalidation against the gauge block.
- **Quarterly:** lighting replacement based on hours.

6. References

- Maintenance_Workflow_Overview.pdf
- Quality_Reject_Disposition_Policy.pdf
- Shift_Handover_Checklist.pdf
- Line-B_03_Trimming-Station_SOP.pdf
- Production_Lines_Reference.pdf